FEDERAL SPECIFICATION

AMBULANCE

EMERGENCY MEDICAL CARE VEHICLE

This amendment which forms a part of Federal Specification KKK-A-1822A dated April 1, 1980, is approved by the General Services Administration for the use of all Federal agencies.

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Paragraph 1.1.3 First line, change the date "January 1, 1981" to "July 1, 1981."

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Paragraph 2.2 Under Society of Automotive Engineers (SAE), Delete: J555: Truck, truck-tractor, trailer and motor coach wiring.

Add to the bottom of the list:

J1292: Automobile, truck, truck-tractor, trailer and motor coach wiring.

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Paragraph 3.4.2 Delete entirely and substitute:

Paragraph 3.4.2 Temperature conditions. The ambulance vehicle and the exterior mounted equipment and devices shall be capable of normal satisfactory performance in ambient temperatures of -30 degrees to +125 degrees F, when serviced in accordance with the manufacturer's recommendations (see 4.4). Other equipment (including medical) furnished inside the ambulance shall be capable of normal operation after storage in -30 degrees F and +150 degrees F for 24 hours, then operated for one hour at 0 degrees F and 110 degrees F after a 10 hour storage at these temperatures. Type certifications are acceptable when in accordance with 4.3.4.

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Paragraph 3.4.10 Delete entirely and substitute:

3.4.10 Fording. The vehicle shall be capable of three (3) fordings, without stalling the engine, or damage to running gear, while being driven through a minimum of eight inches of water at speeds of five to ten mph for a distance of at least 100 feet. Test under 4.4.4.

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Paragraph 3.5.6 Delete entirely and substitute:

3.5.6 Cab to axle (CA) type I and III vehicles. Cab to axle (CA) dimension of the vehicle chassis shall permit a minimum of 50 percent of the outside body length forward of the rear axle centerline, including the cab to body clearance. Bodies designed with wheel opening shall have the rear wheels centered ± 2 inches within the opening.

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Paragraph 3.6.4.5 Delete entirely and substitute:

3.6.4.5 Cooling system. The engine cooling system shall be a closed, air free liquid state type, with a coolant overflow recovery tank and compensating system. The supplier shall provide the heaviest duty components and maximum size cooling system available from

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the chassis manufacturer applicable to the vehicle offered. The cooling system design shall maintain the engine at safe operating temperatures at all driveable altitudes and grades encountered during on and off road vehicle use. Verification test; the cooling system shall be capable of maintaining a safe engine operating temperature for a period of not less than 40 minutes at sea level in an ambient temperature of 95 degrees F, ± 2 degrees, at the engine speed required to maintain the generating systems as specified in 3.7.6.

Paragraph 3.6.4.6 Exhaust system. After the last sentence in this paragraph, add the following:

"Exhaust tailpipe (outlets) shall not terminate within six inches of the vertical axis of fuel filler opening(s).

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Paragraph 3.7.2 Delete entirely the portion of the paragraph on this page and substitute the following:

3.7.2 Wiring installation. The ambulance body and accessory electrical equipment shall be served by circuit(s) separate and distinct from vehicle chassis circuits. All vehicle wiring shall be copper and conform to all the SAE J1292 requirements and have type GPT thermoplastic or better insulation conforming to SAE J1128. The wiring shall be color coded or permanently marked for identification with easily read numbers and/or letters, and routed in conduit or looms conforming to SAE J562 as applicable. All wiring shall be located in accessible, enclosed, and protected locations and kept at least six inches away from exhaust system components. Electrical wiring and components shall not terminate in the oxygen storage compartment except for the compartment light and switch plunger or trigger device. Wiring necessarily passing through an oxygen compartment shall be routed in a metallic conduit (see 3.11.3). All conduits, looms, and wiring shall be secured to the body or frame with insulated metal cable straps in order to prevent sagging and movement which results in chafing, pinching, snagging or any other damage. All apertures on the vehicle . . .

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Paragraph 3.7.7 Delete the 2nd paragraph and substitute:

"Battery ratings shall conform to SAE J537. Batteries shall be located in a ventilated area, sealed off from occupant compartments, and shall be readily accessible for servicing and removal. When batteries are mounted in the engine compartment, they shall be provided with a heat shield as a safeguard against high underhood temperatures."

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Paragraph 3.7.7.3 Delete entirely and substitute:

3.7.7.3 Internal 12 VDC power, (reference figure 5). The patient compartment shall be furnished with a 12 VDC, 20 ampere capacity, separately protected circuit, with two (2) outlet receptacles. This circuit shall also include a (low voltage drop) "Schottky" diode to isolate medical equipment batteries from any electrical loads that the remainder of the ambulance electrical system may impose. The Schottky diode shall be heat-sink mounted, have an inverse voltage rating of at least 45 volts, and also be rated to carry the maximum short circuit current until the circuit breaker opens. The diode shall be physically located in an accessible location and be electrically connected between the circuit breaker and the "action wall" mounted receptacle. Unless otherwise specified, the receptacles shall be a military type connector of the following generic designation, MS3112E12-3S, or its interchangeable commercial equivalent. The polarity of the connector shall be as follows: Pin A -+12V, Pin B - Ground, Pin C - not used. The receptacles shall be located on a vertical surface of the "action wall". The mating plug attached to the medical equipment shall be an MS3116F12-3P or its interchangeable commercial equivalent. The polarity for the plug shall be the same as above. Two of these unwired plugs shall be furnished and tagged with polarity requirements and shall be connected to the receptacles. (NOTE: These connectors are widely available directly from most major industrial electronics distributors).

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Paragraph 3.7.12 Delete entirely and substitute:

3.7.12 Electromagnetic radiation of electrical and electronic components. In addition to OEM chassis, all electrical components, electronic equipment, and devices used and installed on or in the ambulance, shall be electromagnetic radiation suppressed, filtered, or shielded to prevent interference to radio and telemetry equipment aboard the vehicle and the surrounding area. Such devices and equipment shall be tested and certified to prove that their RFI does not exceed the maximum limits of SAE J551. Type approval test acceptable as per 4.3.4 for 4.4.2.

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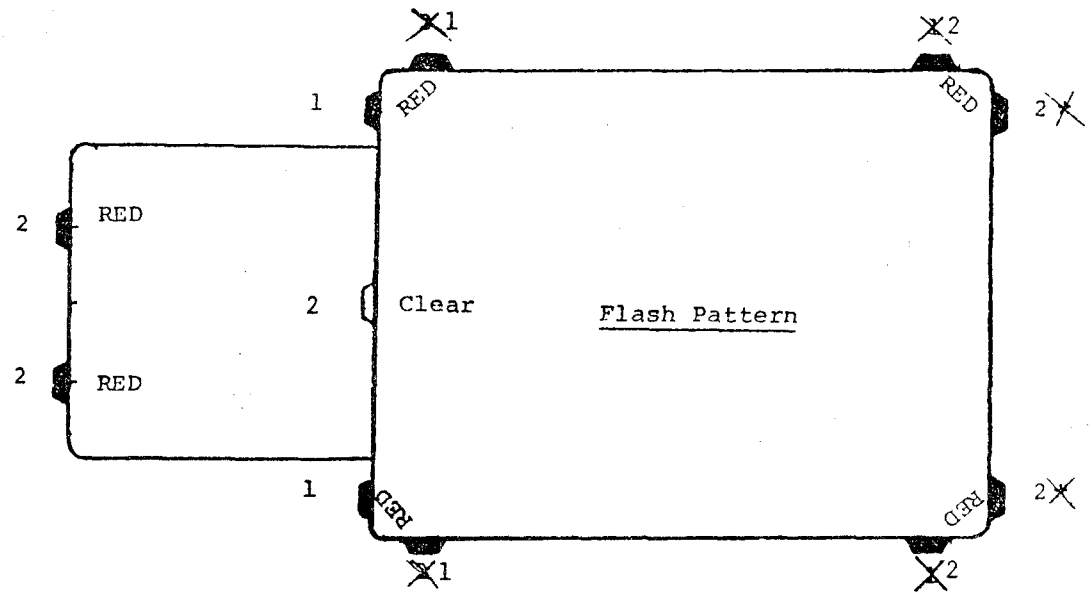
Paragraph 3.8.2 Delete entirely and substitute:

3.8.2 Ambulance emergency lighting. The emergency lighting system must provide the ambulance with 360 degrees of conspicuity for safety during its missions. The system must display highly perceptible and attention-getting signals that function in a modal system, and convey the message in the PRIMARY MODE - "Clear the Right-of-Way" and in the SECONDARY MODE - "Hazard Vehicle Stopped on Right-of-Way". The ambulance standard warning light system shall not impose an electrical load exceeding 40 amperes. Additional auxiliary warning lights, if specified (see 6.2), shall not obscure the light output of the standard warning light system. Auxiliary warning lights furnished shall be separately switched. Any warning devices furnished in addition to the specified system shall be compensated for with reserve or additional generating capacity as required in 3.7.6.

The ambulance standard emergency warning light system shall contain ten (10) fixed red lights and one (1) fixed clear light. These lights shall function in a dual mode system as shown in Table I, and meet the physical and photometric requirements of 3.8.2.1. The upper body warning lights shall be identical and mounted at the extreme upper corner areas of the ambulance body below the horizontal roofline, with the single clear light mounted midway between the two front facing red upper corner lights (see figures 1, 2, and 3). The standard warning lights shall not be obstructed by doors, auxiliary lights, or siren(s). The red "grill" lights shall be located at least 30 inches above the ground but below the bottom edge of the windshield and be laterally separated by at least 24 inches. All warning lights furnished shall be mounted to project their highest effective intensity beams on the horizontal axis.

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Delete, and substitute the following:



1 - indicates lights flashing at the same time
2 - indicates lights flashing 180° out of phase with 1
~~*In secondary mode, upper rear lights may alternately flash.~~
**The "Tertiary Mode" which uses the vehicles 4-way flashers (Directional Signals) should only be used while the vehicle is stationary.

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Paragraph 3.8.2.2 Tenth line down after "mode" insert "and, when specified (6.2), the Day-Night switch".

Paragraph 3.8.2.3 Delete entirely and substitute:

3.8.2.3 Tests, warning light system. The ambulance manufacturer shall measure and record the total current load of the standard emergency warning light system on the vehicle as manufactured, when operated in the mode which draws maximum current. This load current test shall be conducted during the "ambulance's electrical system test" (3.7.6 and AMD Std. 005). The standard warning light system and related components and devices shall be tested, approved, and listed with the American Association of Motor Vehicle Administrators (AAMVA) for conformance to the requirements herein.

Paragraph 3.8.5 Delete entirely and substitute:

3.8.5 Ambulance interior lighting. The basic interior ambulance lighting configuration shall be designed to minimize electrical loads and include: A driver's compartment dome light, instrument panel lights, master switch panel and console light(s), and glove box light. Lighting shall be designed and located so that no glare is reflected into the driver's eyes or his line of vision from switch control panels or other areas that are illuminated while the vehicle is in motion. The patient compartment lighting 3.8.5.1 shall also include stepwell lighting (3.10.12) and control panel lighting.

Paragraph 3.10.2 Delete entirely and substitute:

3.10.2 Cab and body access between compartments, Type 1. The ambulance cab and body bulkheads shall have an aligned window opening of at least 150 square inches for visual and voice check of conditions in the patient's compartment. The window opening shall be an adjustable, transparent, shatterproof panel treated or located in the bulkhead(s) to prevent interferring with the driver's night vision. (See 3.9.9 and 3.10.14).

Paragraph 3.10.4 Delete the third paragraph "Width", and substitute:

"Width: The width of the compartment, after installation of the cabinets shall provide 19 inches ± 6 inches of clear aisle walkway between the secured primary cot and the squad bench or cot."

Paragraph 3.10.6 Delete entirely and substitute:

3.10.6 Ambulance body structure. All parts of the ambulance body and attachments shall be fastened with rust resistant fasteners in a manner which will preclude loosening. Cabinets, benches, partitions, oxygen cylinder holders, guide rails, and cot holders shall be attached to metal tapping plates and/or framing welded to the body structure. These components shall be fastened by welding, bolting, or using tapered tapping screws, at least on 24 inch centers or less as applicable to the component being installed. Self-tapping wood/metal screws or nails shall not be used in assembling the ambulance except for light trim panels and wood flooring.

Vehicles furnished with fiberglass/plastic exterior roof panel shall have the center section reinforced with metal wire screening (see 3.14.3 for a radio antenna ground plane). Ambulance bodies with an extended roof shall have the roof structural members permanently fastened to structual members of the body (welded, bolted, and sealed) to prevent separation in an accident. Drip rail(s) shall be provided over doors of type I and III or around the entire module unit and have drain points at each corner. Body skirt(s) shall not extend more than 3 inches below the vehicle cab/body. On type I and III bodies, when specified (see 3.15.3 - 18), a minimum 1x2 inch protective rub rail on the right and left sides located in the lower third section of the body shall be provided. The body, roof, and panel joints shall be watertight. All openings between the chassis-body and occupant carrying compartments due to alteration or construction shall be sealed, including the bulkhead space between cab and body of type I and III (see 3.9.9).

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Paragraph 3.10.9 Delete entirely and substitute:

3.10.9 Door latches, hinges, and hardware. Door latches, hinges, and hardware furnished by chassis manufacturers shall comply with FMVSS 206. When doors are open, the hinges, latches, and door-checks shall not protrude into the access area. All doors shall have hardware or devices to prevent inadvertent opening and closing. A minimum 6-inch grab handle on the inside of each door, in addition to a door operating handle; door stops to prevent damage to body sides; a handle with latches operable from inside and outside of the body with one external operated lock with key per door opening shall be provided. Hardware shall be chrome plated, stainless steel, or anodized aluminum. Inside door handles shall be designed and placed so they cannot be operated (opening a door) when accidentally hit or used as a grab handle. Ambulance body, side and rear door hardware installed by the ambulance body manufacturer shall be tested to prove installation meets or exceeds the requirements of AMD Standard 002 -Body and door retention components tests.

Paragraph 3.10.15 Delete entirely the portion on this page and substitute:

3.10.15 Partition for Type II and III vehicles. A full height and width partition or bulkhead (with or without compartments) having roll bar characteristics and an opening with a door, shall be placed between the driver and patient's compartment. This partition shall be located directly behind the driver and companion seats when in the . . .

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First paragraph, delete entirely and substitute:

rearmost position. The partition shall be secured at the sides, ceiling, and floor, by welding or bolting to tapping plates. A partition opening at least 17 inches wide and 46 inches high shall provide an aisle between the compartments. The door shall have at least a 150 square inch transparent shatterproof viewing panel in the center section at the driver's eye level. The door shall be securable with a self-latching device in the open and closed positions from the driver's side (see 3.10.2).

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Paragraph 3.11.3 Delete entirely and substitute:

3.11.3 Storage compartments and cabinets design. Storage cabinets, drawers, and kits shall be easily opened but shall not come open in transit. For rapid identification of contents, medical supply cabinets above the litter patients shall have shatterproof transparent sliding doors provided with a finger pull opening or recessed metal cups or equal. Storage compartments shall be divided into sections, shelves shall be adjustable, drawers shall be marine style slide or tilt, and all shall be removable. Cabinet compartment doors and drawers, sliding or hinged, shall automatically latch or be fitted with friction holding devices when in a closed position. Side cabinet shelves shall be no more than 12 inches in depth when located above the vehicle belt level. Storage compartments, cabinets, and support equipment area interior surfaces shall be finished in accordance with 3.10.17. Cabinets shall be firmly anchored (bolted or welded) to tapping plates of the body structure (3.10.6). Tops of the cabinets and shelves shall be bordered or surrounded by a lip of not less than 1/2 inch in height. Storage for the main oxygen cylinder (see 3/12/1) shall be accessible for replacement from an outside position. The oxygen compartment shall be provided with at least a nine (9) square inch louvered device located near or at the top of the compartment, permitting any leaking oxygen gas to dissipate/vent to the outside of the ambulance. Oxygen cylinder compartment shall not be utilized for storage of any other equipment. Any wiring and electrical devices within this compartment shall comply to 3.7.2. Oxygen cylinder(s) shall be mounted with a restraining device(s), required for the crashworthiness tests of AMD Standard 003 Oxygen Tank Retention System.

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Paragraph 3.11.9 Delete entirely and substitute:

3.11.9 IV holders for intravenous fluid containers. Two near flush style, IV ceiling holders or hooks with strapping device to tie and control IV bags/bottles shall be provided. The ceiling holders shall be located adjacent to the side wall, at the head of the primary patient and one at the head of the secondary patient's cot (squad bench). When specified in 3.15.4, code M2, a detachable type, rigid telescoping IV pole and holder, with a 52 inch minimum height, when extended, shall be provided the style 1 cot. It shall be mounted on the left side at the front end of the cot.

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Paragraph 3.13.1 After the last sentence add:

"Connecting hoses for the heating and air conditioning systems shall be supported at least every 12 inches by rubber insulated metal clamping devices."

Paragraph 3.13.4 and 3.13.5 Fourth line down, delete "(floor to ceiling)".

Page 40

Paragraph 4.3.1 Second line, change "January 1, 1981" to "July 1, 1981".

Page 41

Paragraph 4.4.2 Line "h" change to read:

"Electromagnetic radiation of electrical/electronic devices tests. Type approval acceptable."

Page 43

Paragraph 6.2 Ordering data, item (g) Wheels, after: "desired", delete: "if Type I".

Page 47

Paragraph 6.9 Changes and amendments, delete: "Federal Supply Service, Automotive and Tools Division (FREA-A), and substitute: "Transportation and Public Utilities Service (TCE)".

Change Civil Coordination & Preparing Activity from: "GSA-FSS (FREA)" to "GSA-TPUS-TCE" in lower right side of page.

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